Complementary disclosures in financial networks

Accounting geometry, prior-weighted uncertainty and bounded-error reports

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Abstract. Balance-sheet totals need not determine outside-creditor losses after a financial network clears a known shock. We study the residual range of this loss after exposure reports, rather than the welfare consequences of disclosure. Two mechanisms must be distinguished. On a coupled accounting simplex, an affine loss exhibits all-or-nothing complementarity under a restricted report menu: one interbank layer suffices for arbitrarily many jointly necessary reports. For this affine class, standard linear-programming duality gives an exact menu formula, an identification criterion and a bounded-error extension. In a different, layered family, loss contains a product of hidden allocation signs. We determine its uncertainty relative to the shock and its dependence on the ratio of outside to interbank debt. For independent priors used to weight full feasible fibres, we give the exact expected width and a submodularity criterion valid also for degenerate priors; cancellation of zero factors would give a false criterion. We derive quantile widths, exhibit different optimal packages under worst-case and expected-width criteria, and calculate heterogeneous report-error effects. The treatment consolidates an unpublished companion, identifies standard ingredients and supplied-review contributions, and distinguishes written proofs, exact finite checks, solver answers and historical records. The constructions are synthetic; they imply neither prevalence in actual networks nor a welfare-optimal disclosure policy.

1. Question and contribution

A supervisor knows each bank's total interbank liabilities and assets, outside debt and post-shock external assets, but not its bilateral liabilities. Which reports identify outside-creditor loss after proportional clearing, and which combinations are useful?

The question has three ingredients: an accounting-feasible set, a report menu, and an uncertainty criterion. Omitting any can misidentify the source of complementarity. Nonlinear loss is not needed for every example: accounting identities can couple affine quantities, and a menu can omit a directly informative report.

This paper is self-contained. Section 2 supplies clearing, identification and selection. Section 3 treats the accounting-simplex mechanism and gives an exact affine-menu characterisation. Section 4 studies the layered product mechanism. Sections 5 and 6 determine prior-weighted and bounded-error widths. Section 7 separates computational evidence by assurance level. Appendix A recovers the earlier six-bank aggregate calculation.

The additions are quantitative and model-specific, not a new general theory of informational complementarity or LP duality. Their strongest significance limitation is that they concern deliberately constructed networks and stipulated menus. Their useful output is exact diagnostics: what a menu identifies, how much uncertainty remains, when the criterion changes the optimal package, and how precision changes the answer.

1.1 Relationship to earlier work

The supplied unpublished working note, *Optimal exposure disclosure under clearing uncertainty* (Anonymous, 2026), already contains the pair formulation, witness-cut selection, six-bank example, layered construction at a particular parameter value, single-aggregate optimum and robustness variants. It was not located as a public Evidence Press release during this revision. We consolidate the required arguments instead of directing readers to an unavailable published companion. The original note is retained as a historical source, not a separately verified current release.

Ingredient	Relationship to prior material
Proportional clearing and regularity	Eisenberg and Noe (2001); specialised proof below
Pair width; midpoint error equal to half-width	Scalar optimal recovery; supplied companion
Separating excessive-loss pairs	Test-cover/hitting-set formulation; companion specialisation
Abstract high-order complementarity	XOR signals in Chen and Waggoner (2016, §5.1)
Loss–shock bound	Consequence of loss conservation
One-layer affine construction	Supplied review; proved with menu qualifications
Affine-simplex formula	Standard LP duality specialised to disclosure
Layered family	Companion construction with general debt ratio
Prior and noise formulas	Quantitative analysis, corrected zero-factor boundary and heterogeneous errors

Our objective differs from decision-theoretic value of information. Chen and Waggoner (2016) analyse expected decision utility; our expected criterion averages full accounting-feasible interval widths. Ahn, Chen and Kim (2024) study partial information and worst-case shock propagation, including several kinds of known bank and link information. Their risk-quantification problem differs from choosing a finite report package to minimise fixed-shock loss diameter on every fibre. Hu, Mitchell and Tompaidis (2024) likewise consider robust financial networks consistent with aggregate liabilities and assets, making their framework a close benchmark rather than a generic adjacent citation. The comparison here is limited to their publisher abstract, not a theorem-level priority assessment. Feinstein et al. (2018) analyse clearing sensitivity to liability perturbations; our object is finite report-consistent ambiguity, not a local derivative. These distinctions delimit our contribution without claiming exhaustive exclusion of earlier network-information work.

All monetary quantities use a common arbitrary unit. All networks constructed here are synthetic.

2. Clearing, feasible fibres and report selection

2.1 Model and well-posedness

There are $n \geq 1$ banks. The nominal liability $L_{ij} \geq 0$ goes from bank i to bank j, with $L_{ii} = 0$. Known totals satisfy

$$\sum_j L_{ij} = r_i, \qquad \sum_i L_{ij} = a_j.$$

Each bank owes $b_i > 0$ outside and holds post-shock external assets $x_i \geq 0$. Put $\bar{p}_i = r_i + b_i$. The nonempty feasible set $\mathcal{F}$ consists of matrices with these totals, specified zeros and any closed

edge bounds. It is a compact polytope. Clearing satisfies

$$p_i = \min\left\{\bar{p}_i,\ x_i + \sum_j L_{ji}\frac{p_j}{\bar{p}_j}\right\}. \tag{1}$$

The right side is a contraction of the payment box in the 1-norm with constant at most $\gamma = \max_i r_i/(r_i + b_i) < 1$. It therefore has a unique fixed point. The same constant works on $\mathcal{F}$, implying continuity in L by the usual fixed-point perturbation bound with denominator $1 - \gamma$. For $u_i = p_i/\bar{p}_i$, outside-creditor loss is

$$\ell(L) = \sum_i b_i(1 - u_i). \tag{2}$$

The model excludes bankruptcy costs, fire sales and endogenous reactions to disclosure.

The finite menu contains affine reports $Q_q(L)$, $q = 1, \ldots, d$, with costs $c_q \geq 0$. For selected reports S and feasible value y, define

$$\mathcal{F}(S, y) = \{L \in \mathcal{F} : Q_S(L) = y\}, \qquad w(S, y) = \max_{\mathcal{F}(S,y)} \ell - \min_{\mathcal{F}(S,y)} \ell.$$

The extrema exist on each nonempty compact fibre.

2.2 Three uncertainty criteria

The worst-case width is

$$R(S) = \sup_y w(S, y) = \max_{\substack{L,M\in\mathcal{F}\\ Q_S(L)=Q_S(M)}} [\ell(L) - \ell(M)]. \tag{3}$$

The pair-feasible set is compact and nonempty. On a scalar loss interval, its midpoint minimises worst absolute estimation error, giving error $w/2$. Thus $R/2$ is worst-case recovery error.

A prior π can weight likely reports:

$$W_\pi(S) = \mathbb{E}_\pi[w(S, Q_S(L))]. \tag{4}$$

The fibre in (4) is the full accounting-feasible fibre, not posterior support. The prior supplies design weights while the delivered interval protects against every accounting-feasible network. This suits a prior used for planning but not trusted to exclude alternatives. It differs from Bayesian posterior uncertainty; a degenerate prior still does not remove unreported alternatives from w.

The $1 - \alpha$ quantile criterion is the least z with $\pi\{w(S, Q_S(L)) \leq z\} \geq 1 - \alpha$, for $0 < \alpha < 1$.

Write $G_V(S) = V(\emptyset) - V(S)$. This gain is monotone. Reports are substitutes under V when G_V is submodular: a report's marginal benefit cannot increase as the held set grows. A violation is complementarity. A specified m-report menu is *all-or-nothing* when all proper subsets have zero gain and the full set has positive gain. These properties concern the feasible set, menu and criterion jointly.

2.3 General lemmas

Lemma 1 (zero worst-case gain). For compact $\mathcal{F}$ and continuous ℓ, Q, $R(S) = R(\emptyset)$ exactly when a fibre of Q_S contains a global maximiser and a global minimiser of ℓ.

Proof. A maximal pair exists by (3). Its gap equals the global range exactly when both global endpoints are attained. □

Lemma 2 (loss–shock bound). If $x^\circ \geq x$ and every bank pays fully at x°, then for $\sigma = \sum_i (x_i^\circ - x_i)$,

$$0 \leq \ell(L; x) \leq \sigma. \tag{5}$$

Proof. Let $h_i = x_i + \sum_j L_{ji} u_j$ and $e_i = (h_i - \bar{p}_i)^+$. Cancelling interbank transfers gives $\ell = \sum_i b_i - \sum_i x_i + \sum_i e_i$. Payments, cash and equity are monotone in x. Since $\ell(L; x^\circ) = 0$, $\ell(L; x) = \sigma + \sum_i e_i(x) - \sum_i e_i(x^\circ) \leq \sigma$. Nonnegativity follows from $u_i \leq 1$. □

This useful computational bound follows from classical conservation, explicitly discussed by Visentin, Battiston and D'Errico (2016, §3.2 and Supplementary Information §2.2). Our constructions approach ratio one. Inequality (5) alone does not establish universal nonattainment of equality.

Lemma 3 (product modules). For disconnected modules with product feasible set and module-local reports,

$$w(S, y) = \sum_k w_k(S_k, y_k), \quad R(S) = \sum_k R_k(S_k), \quad W_\pi(S) = \sum_k W_{\pi_k, k}(S_k). \tag{6}$$

The last equality holds for **any joint prior** with marginals π_k.

Proof. Fibres are products and losses sum. Maxima and minima separate. Maximise over product report ranges for R, and use linearity of expectation for W. No independence assumption on the prior is needed. □

2.4 Exact selection as pair separation

For tolerance ε, each pair with $\ell(L) - \ell(M) > \varepsilon$ has distinguishing set $D(L, M) = \{q : Q_q(L) \neq Q_q(M)\}$. Formula (3) says $R(S) \leq \varepsilon$ exactly when S hits every such set. This is a scalar loss-tolerance specialisation of test cover.

A witness-cut algorithm solves a cost-minimising hitting-set master over collected sets, then asks whether its selected reports admit another excessive-loss pair. A verified pair adds a cut. An empty distinguishing set proves menu infeasibility. UNSAT from a complete exact oracle certifies optimality, since all previous cuts are necessary.

There are at most 2^d distinguishing sets, and each newly found one is not hit by the current solution. This proves termination with a complete exact oracle, not a practical bound for nonlinear queries or combinatorial selection. UNKNOWN is not infeasibility or optimality. Real-algebraic SAT models need exact validation of their own; they are not automatically rational certificates.

3. Accounting geometry and the report menu

3.1 One-layer affine construction

The following construction was suggested in the supplied review.

Proposition 1 (affine loss, jointly necessary menu). For every $m \geq 1, b_s > 0$, a $2(m+1)$-bank network with one interbank layer, fixed totals and an m-report menu N satisfies

$$\ell(t) = b_s + t_0, \qquad R(S) = 1 \ (S \subsetneq N), \qquad R(N) = 0. \tag{7}$$

The shock is $1 + b_s$.

Proof. Take $k = m + 1$ sources and sinks with a doubly stochastic interbank block. Sources owe b_s outside and initially have assets $1 + b_s$. Remove source 0's assets. Sinks owe one outside and no interbank debt. Sink 0 has external assets zero; the others have one.

The shocked row is any $t \in \Delta = \{t \geq 0 : \sum_{j=0}^{m} t_j = 1\}$: extend it by giving each other source row $(1-t)/m$. Unshocked sources pay fully. Sink j receives $1 - t_j$; only sink 0 has outside shortfall t_0. The menu reports $t_1, \ldots, t_m$, deliberately omitting t_0. The full menu identifies t_0. For a proper subset observed at zero, rows e_0, e_j, for an unreported $j \geq 1$, agree on the reports and differ in loss by one. Lemma 1 finishes the proof. □

The unresolved-loss ratio $1/(1+b_s)$ approaches one at every fixed order without interbank depth. One interbank layer does not mean one defaulting bank layer: sink 0 may default as well as the shocked source. Nonlinear loss or uncertain-recovery multiplication is not necessary on the coupled feasible set. If t_0 is available at the same cost, however, one report identifies loss. The menu restriction is essential.

3.2 Exact affine-simplex menu formula

Proposition 2. Let $t \in \Delta \subset \mathbb{R}^k$, loss $b + c^{\mathsf{T}}t$, and selected affine reports with linear part A. Write $\operatorname{osc}(v) = \max_i v_i - \min_i v_i$. Then

$$\begin{aligned} R(A,c) &= \max_{\substack{t,u \in \Delta \\ A(t-u)=0}} c^{\mathsf{T}}(t-u) \\ &= \min_{\lambda} \operatorname{osc}(c - A^{\mathsf{T}}\lambda) \\ &= 2 \min_{\lambda,\alpha} \|c - A^{\mathsf{T}}\lambda - \alpha \mathbf{1}\|_{\infty}. \end{aligned} \tag{8}$$

Under separate deterministic report-error bounds $\eta_q \geq 0$,

$$R_{\eta}(A,c) = \min_{\lambda} \left[\operatorname{osc}(c - A^{\mathsf{T}}\lambda) + 2 \sum_{q} \eta_q |\lambda_q| \right]. \tag{9}$$

Proof. Constants cancel. A multiplier λ for $A(t-u) = 0$ gives the supremum $\max_{t,u \in \Delta}(c - A^{\mathsf{T}}\lambda)^{\mathsf{T}}(t-u) = \operatorname{osc}(c - A^{\mathsf{T}}\lambda)$. Finite LP strong duality applies to this feasible bounded primal and yields dual attainment. Centring at the midpoint of the extreme entries proves the last equality.

With bounded errors, common observations exist exactly when $|A(t-u)| \leq 2\eta$. The two inequality multipliers have difference λ; the least penalty for a fixed difference is $2\sum_q \eta_q |\lambda_q|$, attained by positive and negative parts. Duality gives (9), including zero tolerances and redundant rows. □

This is a specialisation of standard LP duality, as in Boyd and Vandenberghe (2004, §§5.2.3–5.2.4), not new general duality. No rank condition is needed. Rational primal and dual feasible solutions with equal objective values provide exact certificates.

Consequently:

1. **Identification:** $R(A,c) = 0$ iff $c \in \operatorname{span}\{\mathbf{1}, \text{rows of } A\}$.
2. **Zero gain:** for extreme-index sets I_+, I_- of c, and columns a_i of A, $R(A,c) = \operatorname{osc}(c)$ iff

$$\operatorname{conv}\{a_i : i \in I_+\} \cap \operatorname{conv}\{a_i : i \in I_-\} \neq \emptyset. \tag{10}$$

3. **Coordinate reports:** for nonempty unreported set U,

$$w(S, t_S) = \left(1 - \sum_{i \in S} t_i\right) \operatorname{osc}(c_U), \qquad R(S) = \operatorname{osc}(c_U). \tag{11}$$

Empty or singleton unreported sets give zero width.

For the first claim, zero gap on $\ker A \cap \mathbf{1}^\perp$ implies the span condition because the simplex difference set contains a neighbourhood of zero in $\mathbf{1}^\perp$; the converse is immediate. A maximal-gap pair is supported on the two extreme coefficient sets, giving (10). The residual simplex has mass $1 - \sum_{i \in S} t_i$, giving (11), with worst case at $t_S = 0$.

This characterises widths for a fixed subset in the affine class. It does not make arbitrary menu selection polynomial-time or generic clearing affine.

Proposition 2 permits arbitrary affine coefficients and report matrices, including asymmetric coefficients, redundant reports and alternative menus containing the directly informative exposure. This supplies a restricted-class robustness statement beyond the symmetric example. It is not a theorem that complementarity survives every perturbation of bank balance sheets: such perturbations may change both the feasible domain and the clearing regime.

3.3 Saturation and the product-domain qualification

A nonlinear one-layer example has **uniform** sink slack. Use $k \geq 2$ sources and sinks with the same doubly stochastic block, source outside debt b_s, and complete shock to one source. Each sink owes one outside and holds assets $1/2$. Its shocked row gives

$$\ell(t) = b_s + \sum_{j=1}^{k} (t_j - \tfrac{1}{2})^+, \qquad R(S) = \begin{cases} \frac{1}{2}, & |S| \leq k-2, \\ 0, & |S| \geq k-1. \end{cases} \tag{12}$$

With at least two unreported coordinates and observed values zero, putting the remaining unit on one sink gives excess $1/2$, while spreading it equally gives zero. With $k-1$ reports the last coordinate is known. The full menu has a $(k-1)$-of-k threshold; any fixed $(k-1)$-report submenu is all-or-nothing of order $k-1$.

In contrast, for a product box, separable loss $\sum_j g_j(t_j)$, and coordinate reports,

$$R(S) = \sum_{j \notin S} \operatorname{osc}(g_j), \tag{13}$$

so gain is modular. All three assumptions matter: Proposition 1 has affine loss but a simplex, not a product box. Feasible geometry, menu restrictions and nonlinear transmission are distinct mechanisms.

4. The layered product mechanism

4.1 Construction and debt ratio

Fix $m \geq 1, \beta > 0, b_0 > 0$. Layer j, $j = 0, \ldots, m$, has banks $2j, 2j+1$. Between successive layers the liability block is

$$T(t_j) = \begin{pmatrix} t_j & 1-t_j \\ 1-t_j & t_j \end{pmatrix}, \qquad t_j \in [0,1], \quad j = 1, \ldots, m.$$

There are no other interbank liabilities. Sources owe b_0 outside, intermediates owe β outside, and final banks owe one outside. Before the shock, external assets are $1 + b_0$ at each source, β at each intermediate, and $(0, 1)$ at the final pair. All pay fully. Removing bank 0's assets gives shock $\sigma = 1 + b_0$.

Each interlayer block has unit row and column sums independent of its switch. Sources have interbank column sum zero and row sum one; intermediates have both one; final banks have row sum zero and column sum one. Totals are fixed but not all one at the boundary. Reports are $t_j = L_{2(j-1),2j}$. The parameter β is the **outside-to-interbank debt ratio**, not the outside share; that share is $\beta/(1+\beta)$.

Figure 1: The layered family at $m = 3$. Each interlayer block has unit row and column sums; source and final-layer boundary totals differ as described in the text. Each intermediate layer transfers the fraction $\beta/(1+\beta)$ of its incoming deficit to outside creditors. Uncertain residual allocation determines final-layer loss.

4.2 Exact loss and widths

Theorem A (layered magnitude). Put $c = (1+\beta)^{1-m}$. For every $t \in [0,1]^m$,

$$\ell_m(t) = 1 + b_0 - \frac{c}{2}\left(1 - \prod_{j=1}^{m}(2t_j - 1)\right). \tag{14}$$

For $S \subsetneq N = \{1, \dots, m\}$,

$$w(S, t_S) = c \prod_{j \in S} |2t_j - 1|, \qquad R(S) = c, \tag{15}$$

whereas $R(N) = 0$. If switches are restricted to $[a, 1-a]$, $0 \le a < 1/2$, and $h = 1 - 2a$, then for proper S,

$$w(S, t_S) = c\, h^{m-|S|} \prod_{j \in S} |2t_j - 1|, \qquad R(S) = ch^m, \tag{16}$$

with zero width after full disclosure.

Proof. Bank 0 pays nothing and bank 1 pays fully. The deficit entering layer 1 is $d_1 = T(t_1)^{\mathsf{T}}(1,0)^{\mathsf{T}}$. An intermediate bank receiving deficit $s \in [0,1]$ has recovery $1 - s/(1+\beta)$, outside loss $\beta s/(1+\beta)$, and transmits interbank deficit $s/(1+\beta)$. Thus

$$d_{j+1} = (1+\beta)^{-1} T(t_{j+1})^{\mathsf{T}} d_j, \qquad \|d_j\|_1 = (1+\beta)^{1-j}.$$

These bounds keep banks in the stated linear regime. The common eigenvectors of $T(t)$ are $(1,1)^{\mathsf{T}}, (1,-1)^{\mathsf{T}}$, with eigenvalues $1, 2t-1$. Hence

$$d_m = \frac{c}{2}\left[(1,1)^{\mathsf{T}} + \prod_j (2t_j - 1)(1,-1)^{\mathsf{T}}\right].$$

Intermediate outside losses sum to $1-c$. The first final bank loses the first component of d_m, and the second final bank's unit slack absorbs its deficit. Adding source outside loss b_0 gives (14).

For proper S, the free-sign product ranges over $[-1,1]$, proving (15). On the restricted box it ranges over $[-h^{m-|S|}, h^{m-|S|}]$, proving (16). Endpoint choices attain the maxima. □

The unresolved-loss ratio is

$$\frac{R(\emptyset)}{\sigma} = \frac{(1+\beta)^{1-m}}{1+b_0}. \tag{17}$$

For $b_0 = \beta$ it is $(1+\beta)^{-m}$, exponentially decreasing in m at fixed positive β. For each fixed m and $\varepsilon \in (0,1)$, choosing

$$0 < \beta \le (1-\varepsilon)^{-1/m} - 1$$

makes the ratio at least $1-\varepsilon$. Maintaining that guarantee as m grows requires β of order $1/m$, not a fixed positive value. Ratios approach one in this family but do not attain it at its positive parameters. Proposition 1 reaches the same supremum without depth.

m	$\beta = 1$	$\beta = 1/10$	$\beta = 1/100$
1	0.500	0.909	0.990
2	0.250	0.826	0.980
4	0.063	0.683	0.961
6	0.016	0.564	0.942
10	0.001	0.386	0.905

Table 1. Formula $(1+\beta)^{-m}$, rounded to three decimals, with $b_0 = \beta$. These are mathematical evaluations, not empirical estimates.

5. Priors weight fibres; they do not redefine them

5.1 Expected widths and the zero-factor boundary

Theorem B (corrected prior criterion). Let the switches be independent under π, with $\kappa_j = \mathbb{E}_\pi|2t_j - 1| \in [0,1]$. Then

$$W_\pi(S) = c\prod_{j\in S}\kappa_j \quad (S \subsetneq N), \qquad W_\pi(N) = 0. \tag{18}$$

For $m \ge 2$, expected-width gain is submodular exactly when, for each distinct i, j,

$$\left(\prod_{k\notin\{i,j\}}\kappa_k\right)(1-\kappa_i-\kappa_j) \ge 0. \tag{19}$$

Equivalently, either some $\kappa_j = 0$, or every pair satisfies $\kappa_i + \kappa_j \le 1$. For $m = 1$ submodularity is automatic.

Proof. Formula (18) follows from (15) and independence. For distinct $i, j \notin S$, when $Sij = S \cup \{i,j\}$ is proper, the elementary submodularity difference is

$$G(Si) + G(Sj) - G(S) - G(Sij) = c\prod_{k\in S}\kappa_k(1-\kappa_i)(1-\kappa_j) \ge 0.$$

When $S \cup \{i,j\} = N$, the difference is the left side of (19) times c. These elementary inequalities are necessary and sufficient, because general diminishing-return inequalities follow by adjoining one element at a time. This proves the condition without cancelling any factor.

If all factors are positive, cancel them to obtain the pair-sum conditions. If $\kappa_z = 0$, a pair not containing z has zero prefactor; a pair containing z has nonnegative final factor. Hence every zero-containing vector satisfies (19). □

At $(\kappa_1, \kappa_2, \kappa_3) = (0, 3/4, 3/4)$, gain is submodular despite a pair sum $3/2$. Strict final-two-report complementarity requires both $\kappa_i + \kappa_j > 1$ and $\prod_{k \notin \{i,j\}} \kappa_k > 0$.

Uniform switches have $\kappa = 1/2$, giving equality in completion inequalities, but not necessarily modularity when $m > 2$. Other examples are $3/8$ for Beta$(2, 2)$, $9/16$ for Beta$(1, 3)$, and $2/\pi$ for Beta$(1/2, 1/2)$. Endpoint-supported priors have $\kappa = 1$ and yield the worst-case width set function. This concerns (4), not equality of posterior and worst-case uncertainty.

For a proper unit-cost budget $k < m$, the k smallest κ_j minimise (18). Greedy expected-width selection is therefore optimal in this family, with harmless ties after a zero is selected; no such general guarantee is claimed for clearing networks.

5.2 Quantile corollary

Corollary C. For independent uniform switches and $1 \leq n < m$ observed reports,

$$q_n(\alpha) = c \exp[-\gamma_n(\alpha)], \tag{20}$$

where q_n is the $1 - \alpha$ quantile of realised width and γ_n is the α-quantile of Gamma$(n, 1)$. With no reports $q_0 = c$; with all reports $q_m = 0$.

Proof. The $U_j = |2t_j - 1|$ are independent uniform variables. Thus $-\log \prod_{j=1}^{n} U_j$ is Gamma$(n, 1)$, and $c \prod U_j > q_n(\alpha)$ has probability α. □

This uses the standard product-of-uniforms transformation. After six of at least seven reports the 95th percentile is about $0.073c$: the realised width exceeds this threshold with probability 5%. It is not a 5% probability of a false reported interval. Each realised full-fibre interval retains its accounting guarantee.

5.3 Exact optima can depend on the criterion

Proposition 3. Take a six-bank core with $m = 2, \beta = b_0 = 1$, reports x, y, and two disconnected four-bank single-switch modules scaled to widths $7/20, 1/10$, reports z_1, z_2. With unit costs and budget two, the unique minimax optimum is $\{x, y\}$, width $9/20$. Under independent uniform switch priors, expected-width optima are **both** $\{x, z_1\}$ and $\{y, z_1\}$, width $7/20$.

Proof. Lemma 3 and Theorem B give all possibilities:

Package	Worst-case width	Expected width
$\{x, y\}$	$9/20$	$9/20$
$\{z_1, z_2\}$	$1/2$	$1/2$
$\{x, z_1\}$ or $\{y, z_1\}$	$3/5$	$7/20$
$\{x, z_2\}$ or $\{y, z_2\}$	$17/20$	$3/5$

Monotonicity excludes improvement by a smaller package. □

The worst case gives no credit for one core report; expectation credits its typical fibre reduction. Neither criterion is universally preferable.

6. Bounded reporting error

Each reported value has arbitrary error bounded by $\eta_q \geq 0$. Two networks share a possible observation exactly when

$$|Q_q(L) - Q_q(M)| \leq 2\eta_q \qquad (q \in S), \tag{21}$$

because their error intervals intersect. Formula (3) with this constraint gives noisy width. It models bounded additive uncertainty, not fixed rounding bins or random measurement errors.

For layered switches put $\delta_j = \min(4\eta_j, 2)$, $D = \max_j \delta_j$. The factor four converts pair differences in t_j into differences of $2t_j - 1$.

Lemma 4 (heterogeneous product gap). For $m \geq 1$, $\delta_j \in [0, 2]$, and $a, a' \in [-1, 1]^m$ satisfying $|a_j - a'_j| \leq \delta_j$, the maximal product gap is

$$F(\delta) = \begin{cases} 1 - \prod_j (1 - \delta_j), & D \leq 1, \\ D, & D \geq 1. \end{cases} \tag{22}$$

The two expressions agree at $D = 1$.

Proof. If $D \leq 1$ and a coordinate pair has opposite signs, the gap is at most $|a_j| + |a'_j| \leq \delta_j \leq 1 - \prod_i (1 - \delta_i)$. Otherwise take each pair's absolute values, preserving the modulus of the product gap, and suppose $\prod a \geq \prod a'$. For fixed a, the least admissible a'_j is $(a_j - \delta_j)^+$. If some $a_j < \delta_j$, the gap is at most $a_j \leq D$, bounded by the stated expression. Otherwise $\prod_j a_j - \prod_j (a_j - \delta_j)$ is nondecreasing in every coordinate, since its partial derivatives are nonnegative. Its maximum occurs at $a = \mathbf{1}$, attained with $a'_j = 1 - \delta_j$.

For $D \geq 1$, common signs in every coordinate give products of common sign and gap at most one. Otherwise an opposite-sign pair bounds the gap by $\delta_j \leq D$. Set $a = \mathbf{1}$, change one coordinate of a' to $1 - D$ where $\delta_j = D$, and leave the rest one. This attains D. □

Theorem D. With $c = (1 + \beta)^{1-m}$,

$$R_\eta(N) = \frac{c}{2} F(\delta), \qquad R_\eta(S) = c \quad (S \subsetneq N). \tag{23}$$

Proof. Formula (14) makes the loss gap $c/2$ times the product gap. Constraints (21) are those of Lemma 4; clipping at two changes no feasible pair. A proper subset already has a noiseless pair attaining global width c, which no noisy pair exceeds. □

Equal errors $0 \leq \eta \leq 1/4$ recover

$$R_\eta(N) = \frac{c}{2}[1 - (1 - 4\eta)^m]. \tag{24}$$

If every $\eta_j \leq 1/4$, the residual fraction is

$$\frac{R_\eta(N)}{c} = \frac{1}{2}\Big[1 - \prod_j (1 - 4\eta_j)\Big] \leq 2 \sum_j \eta_j.$$

For fixed m, it is asymptotic to $2\sum_j \eta_j$ as the nonzero error vector tends to zero. This is not the coarse-error formula. If some $\eta_j \geq 1/2$, full disclosure leaves width c.

At $m = 2, \beta = 1, \eta = 1/100$, the residual is $49/2500 = 0.0196$, or 0.0392 of initial width. Eight reports each with $\eta = 1/1000$ leave about 0.0158 of initial width. These are model calculations, not practical reporting recommendations.

7. Computational evidence and reproducibility boundaries

Written proofs establish continuum statements. Finite arithmetic checks test implementations and selected consequences; solver checks test formulations without supplying the closed form. Neither constitutes proof-assistant verification.

7.1 Supplied audit and current replay

A current producer-side replay of the supplied pure-arithmetic audit passed 1,740 layered clearing checks, 1,260 endpoint-width checks, 336 realised-width checks, 239 saturation checks, 775 prior-parameter vectors, 12 noisy configurations and 400 cyclic loss–shock cases. The prior grid contains 302 counterexamples to the original zero-cancellation criterion; the corrected condition passes. These are finite tests, not complete parameter coverage. This supplied-audit replay used normal Python mode.

The review separately supplies 25 quantile comparisons and a one-layer script recording 441 exact clearing evaluations, 360 endpoint checks and 12 full-matrix solver queries. These supplied counts are distinct from current pure-arithmetic replay counts. "Review" identifies supplied materials, not verified external peer review, human reviewer identity or independent reproduction.

A separate original release checker, **code/verify_extensions.py**, passed in normal and optimised Python modes. It checks 70 exact affine primal/dual certificates, the 775 prior vectors, 155 heterogeneous-error configurations across 11,154 product-polytope vertices, 375 one-layer rational clearing cases and 78 proper-menu endpoint pairs. Seven negative controls reject malformed certificates or invalid error-domain reasoning. Receipts are **logs/pypi-extensions-normal.json** and **logs/pypi-extensions-optimized.json**. These are finite exact checks supporting, not replacing, the proofs.

7.2 Historical random-network study

The original package explores 32 three-report feed-forward instances, six four-report instances and six cyclic instances. Its records report 24 completed and eight unresolved three-report cases, four completed four-report cases, and one completed cyclic case. The old solver handling did not uniformly retain separately replayable exact rational witnesses for every SAT result. Those counts and brackets are **historical records**, not a freshly certified prevalence sample.

The three-report generator requests $2n + 2$ edges but is capped by $n(n-1)/2$; five-bank runs have ten actual edges, not twelve. A bracket with lower endpoint zero does not prove zero width. The legacy selection target was half the *upper endpoint* of the initial bracket, not half its true width; at tolerance $\sigma/64$ this can overshoot by $\sigma/128$. Longest-query timings omit some selection work, and wall times can include unsuccessful calls. Completion-conditioned results are not population frequencies; small unmatched samples do not establish cycles as the cause of solver difficulty. The original headline prevalence and timing claims are therefore not repeated as current conclusions.

7.3 Fresh retained-witness study

A fresh run reused the 44 historical *input specifications*, not their answers, with the repaired code, Z3 5.1.0, two workers, a 40-second timeout per solver call and bracket tolerance $\sigma/64$. Every instance and every query was retained in **results/fresh-full/**. The aggregation in **results/fresh-study-aggregation.json** gives:

Input group	Inputs	Width-complete	Unresolved	Zero shock
DAG, 5 or 6 banks, 3 reports	16	14	2	0
DAG, 8 or 10 banks, 3 reports	16	10	5	1
DAG, 6 banks, 4 reports	6	4	2	0
Cyclic, 4–6 banks, 3 reports	6	1	5	0
Total	44	29	14	1

Table 2. Fresh finite run. Width-complete means every requested width bracket completed, not that all conceivable questions about that instance were resolved. The zero-shock instance was identified separately and required no solver queries. All eight five-bank DAG inputs have ten actual allowed edges, although twelve were requested.

Among the 29 width-complete instances, 28 have a strictly positive witnessed initial width. The remaining instance has a separate UNSAT answer to the strictly-positive-gap query; its zero-width classification does not rest on a zero lower bracket. Four of the 28 positive-width instances have bracket-certified violations of submodularity. These observations are conditional on these inputs, timeout and completion rule; neither the proportion nor its change from the historical run estimates population prevalence.

Selection was attempted only for those 28 positive-width instances, with the implemented target $R_{0,\mathrm{upper}}/2$. Seven returned an optimal package and 21 returned that no subset of the supplied menu meets this target; none of these selection attempts remained unresolved. The target can exceed half the true initial width by at most $\sigma/128$. This is menu infeasibility, not inconsistency of the underlying network constraints.

Accounting-redundancy queries screened 91 reports across the 29 width-complete instances: 24 were fixed by the accounting constraints, 67 had exact witnessed variation, and none of these screens was unresolved. Of 65 reports whose singleton gain was certified at most $\sigma/64$, 24 were accounting-fixed and 41 were accounting-variable. The other 26 reports were variable but were not certified negligible; this alone is not a positive-gain certificate. The screen does **not** establish that a variable report affects loss, lies in the loss-relevant shocked component, or becomes useful only jointly. Those stronger distinctions remain outside this finite audit. In particular, negligible singleton gain is not automatically evidence of complementarity.

The 1,878 retained queries contain 657 exactly checked rational witnesses, 1,207 solver UNSAT answers and 14 UNKNOWN answers; no solver-only algebraic SAT arose in this run. Replaying all retained rational witnesses and metadata checks passed in normal and optimised Python modes, with receipts **logs/assurance-final-normal.json** and **logs/assurance-final-optimized.json**. These replays do not turn UNSAT answers into independently checked unsatisfiability certificates. The verifier also exercises the algebraic-SAT and UNKNOWN branches through explicit controls.

The two-worker elapsed time was 3,006.9 seconds (50 minutes 6.9 seconds), measured between the run's UTC start and finish. The longest solver call was 40.6744 seconds; the longest query wall time, including extraction and checking overhead, was 40.6864 seconds. Both maxima include all retained stages, including selection and redundancy, and the elapsed time includes unresolved calls. A requested solver timeout is not an exact wall-clock cap. The unmatched groups and small cyclic sample do not identify a causal runtime effect of cycles.

7.4 Certificate semantics

For a query asking whether a gap exceeds a threshold, an exact validated witness supplies an attained lower bound. UNSAT supplies a solver-relative upper bound. SAT without a separately checked exact witness is solver-only evidence and does not advance a witness-certified lower

bound. UNKNOWN preserves the bracket and is not infeasibility, including at initial feasibility or selection.

Reusable witnesses record both matrices, recoveries, gap, selected reports and equality or error constraints, instance data, threshold, solver version and checksums. Algebraic models require exact algebraic validation or an explicit unverified label. In the affine-simplex class, rational primal and dual feasible solutions of equal objective certify a width without nonlinear solving.

Final release receipts identify fresh checks and hashes. Historical outputs remain separately labelled; parsing an old JSON does not promote it to verified evidence. Reproduction is producer-side unless independent external reproduction is specifically documented.

8. Interpretation and limits

There is no unconditional "value of a report." Changing the uncertainty criterion can change an optimal package. Our prior-weighted criterion retains full feasible fibres; posterior-support widths answer a different question.

On a simplex, restricted menus make affine measurements jointly sufficient. On a product domain, layered loss creates a multiplicative interaction with calculable magnitude and precision sensitivity. Saturation supplies another example. These do not exhaust clearing mechanisms. Proposition 2 completely characterises width only in its stated affine-simplex class.

Depth is not necessary for arbitrary-order complementarity. Nor is the saturation example's half-deficit range a universal one-layer bound: Proposition 1 disproves that interpretation. Open questions concern realistic accounting constraints and menus, collection versus transmission costs, and scalable certified bounds for nonlinear clearing.

These are identification results, not welfare or actual financial-stability conclusions. No actual bank data are analysed. This is an unrefereed candidate with written arguments and finite checks at stated scopes, not a formally verified theorem collection or independently reproduced empirical study.

Appendix A. The six-bank aggregate calculation

This content is inherited from the supplied companion. Set $m = 2, b_0 = \beta = 1$, with switches x, y. Then

$$\ell(x, y) = \frac{7}{4} + \frac{1}{4}(2x - 1)(2y - 1).$$

Each individual exposure has zero guaranteed gain; both identify loss. A different one-report menu halves initial width. For $s = x + y$,

$$w(s) = \frac{1}{4}\min(s, 2 - s)^2, \qquad 0 \le s \le 2,$$

with worst case $1/4$, versus initial $1/2$. On $0 \le s \le 1$, xy ranges from zero to $s^2/4$; reflection gives the other half.

This is optimal among single nonconstant affine scalar reports on the square. Reflecting either coordinate preserves absolute loss gaps; exchanging coordinates is a symmetry. A report using both can be normalised to $x + ty$, $0 < t \le 1$. On its fibre $x + ty = 1$, the variable part of twice the loss, up to sign, is

$$g_t(y) = (1 + t)y - 2ty^2, \qquad 0 \le y \le 1.$$

Its minimum is zero. For $t \le 1/3$, its maximum is $1 - t$, giving width at least $1/3$. For $t \ge 1/3$, its maximum is $(1 + t)^2/(8t)$, giving width at least $(1 + t)^2/(16t) \ge 1/4$. A report using only

one coordinate leaves width 1/2. Hence $x + y$ is optimal. This assumes exact real reports and ignores aggregate collection costs.

The companion's greedy failure adds a disconnected one-report module of width $0 < \delta < 1/2$ to this core with budget two. Greedy worst-case gain takes the decoy first; one remaining report cannot complete the core, so gain is only δ. The optimal core pair gains 1/2. The ratio 2δ tends to zero. This concerns gain under that greedy rule, not residual-width approximation or all algorithms.

Provenance and availability

The canonical source is this Markdown manuscript. The companion is retained as the historical unpublished file **provenance/companion-2026-10-13.md**; its exact original hash and input identity are recorded in the package's provenance record. Retention does not endorse every historical computational claim.

The revision and checks are producer-coordinated. The one-layer construction is credited to the supplied review. No human contribution, external peer-review status or exhaustive novelty determination is inferred from those materials. The examples contain no participants or identifiable institution data. Code and verification records accompany the release at stated scopes.

References

Ahn, D., Chen, N., & Kim, K.-K. (2024). Robust risk quantification via shock propagation in financial networks. *Operations Research, 72*(1), 1–18. https://doi.org/10.1287/opre.2020.0722

Anonymous. (2026, October 13). *Optimal exposure disclosure under clearing uncertainty* [Unpublished working note supplied with the candidate; original archived in the release package].

Boyd, S., & Vandenberghe, L. (2004). *Convex optimization*. Cambridge University Press. https://web.stanford.edu/~boyd/cvxbook/

Chen, Y., & Waggoner, B. (2016). Informational substitutes. In *Proceedings of the 57th Annual IEEE Symposium on Foundations of Computer Science* (pp. 239–247). Full version: arXiv:1703.08636v1. https://arxiv.org/abs/1703.08636

De Bontridder, K. M. J., Halldórsson, B. V., Halldórsson, M. M., Hurkens, C. A. J., Lenstra, J. K., Ravi, R., & Stougie, L. (2003). Approximation algorithms for the test cover problem. *Mathematical Programming, 98*, 477–491. https://doi.org/10.1007/s10107-003-0414-6

Eisenberg, L., & Noe, T. H. (2001). Systemic risk in financial systems. *Management Science, 47*(2), 236–249. https://doi.org/10.1287/mnsc.47.2.236.9835

Feinstein, Z., Pang, W., Rudloff, B., Schaanning, E., Sturm, S., & Wildman, M. (2018). Sensitivity of the Eisenberg–Noe clearing vector to individual interbank liabilities. *SIAM Journal on Financial Mathematics, 9*(4), 1286–1325. https://doi.org/10.1137/18M1171060

Hu, F., Mitchell, D., & Tompaidis, S. (2024). Robust financial networks. *Operations Research, 72*(5), 1827–1842. https://doi.org/10.1287/opre.2022.0272

Micchelli, C. A., & Rivlin, T. J. (1977). A survey of optimal recovery. In C. A. Micchelli & T. J. Rivlin (Eds.), *Optimal estimation in approximation theory* (pp. 1–54). Springer. https://doi.org/10.1007/978-1-4684-2388-4_1

Visentin, G., Battiston, S., & D'Errico, M. (2016). *Rethinking financial contagion* [Preprint]. arXiv:1608.07831v1. https://arxiv.org/abs/1608.07831

de Moura, L., & Bjørner, N. (2008). Z3: An efficient SMT solver. In *Tools and algorithms for the construction and analysis of systems (TACAS 2008)* (pp. 337–340). https://doi.org/10.1007/978-3-540-78800-3_24